

1. Project Information

Program	Microbial/CSP 2013
Seq Proj ID	1107807
Sequencing Project Name	Clostridium fallax DSM 2631

2. Read Statistics

Illumina Std PE Statistics

File name	10789.1.179355.TCGCTGT-AACAGCG.anqpht.fastq
Library	BBOPS
Number of reads	8,789,838
Read type	2x149 bp

3. Read QC Results

The following are the results of reads screened against contaminants (human, dog, cat) and artifact sequences. Pairs of matching reads were subsequently subsampled to 10,000,000 reads.

Illumina Std PE Read Filter Statistics

Description	Num Reads	Pct Reads
Input	8,789,838	100
Total removed	0	0.0
Total remaining	8,789,838	100.0

The following are the results of reads screened against potential reagent and process contaminants but were not removed from the dataset.

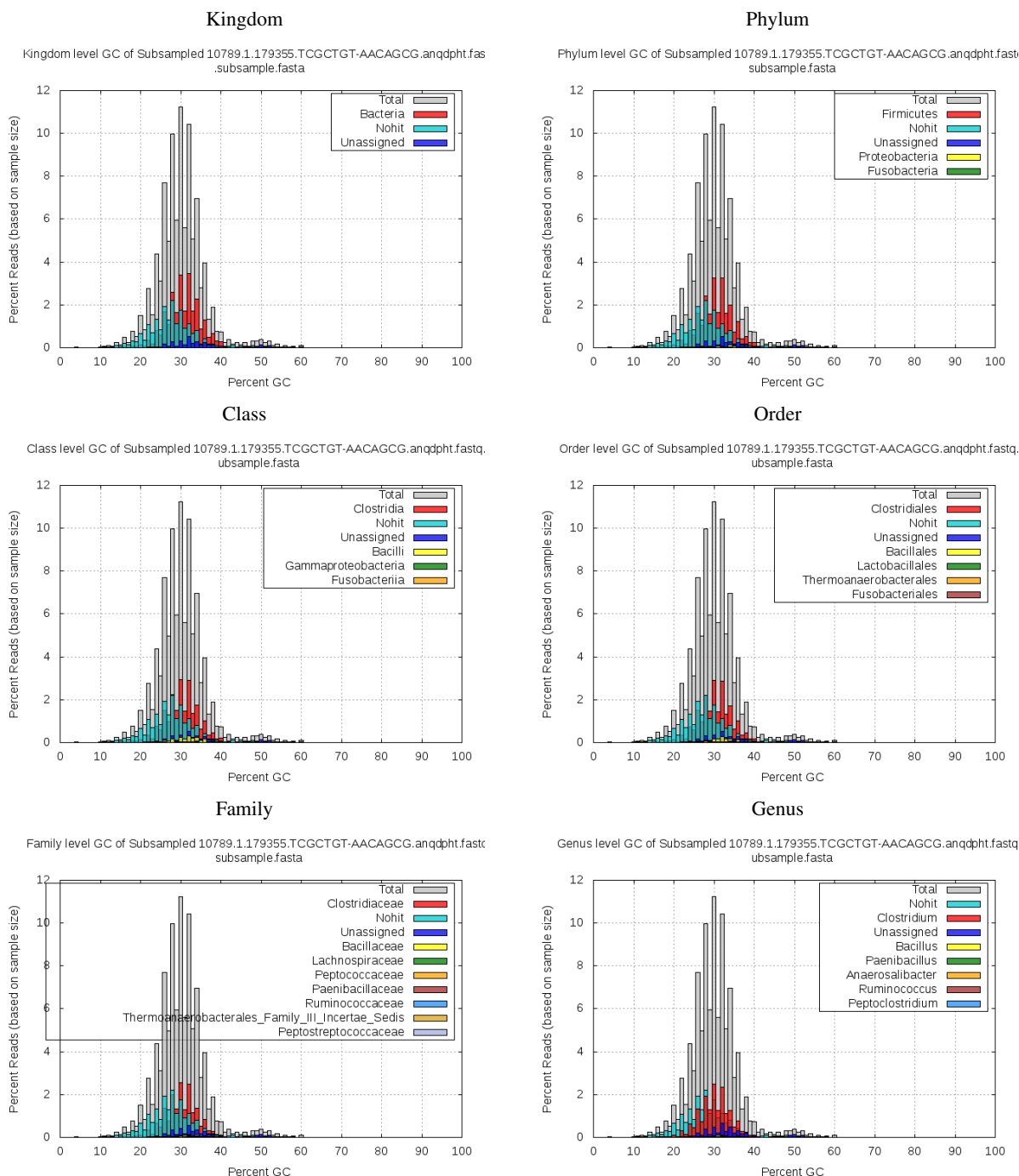
Illumina Std PE Contamination Identification Statistics

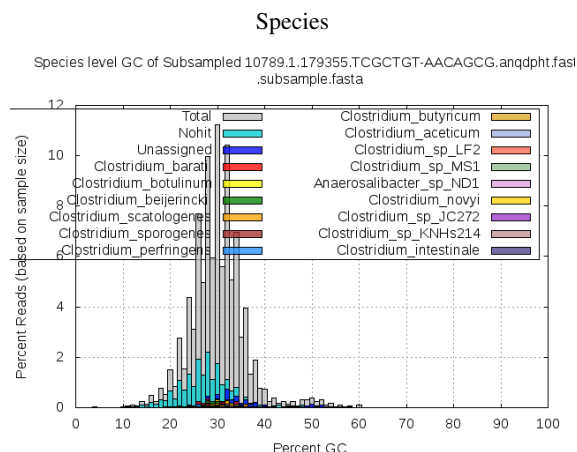
Description	Num Reads	Pct Reads
Input	8,789,838	100
Contam identified	22	0.0

List of Contaminants Identified

Description	Num Reads	Pct Reads
<i>Ralstonia</i>	14	0.00
<i>Cupriavidus</i>	4	0.00
<i>Pseudomonas</i>	4	0.00

GC histogram of the reads subsampled to 10k, overlaid with GC of hits based on LAST, shown for different taxonomic levels.



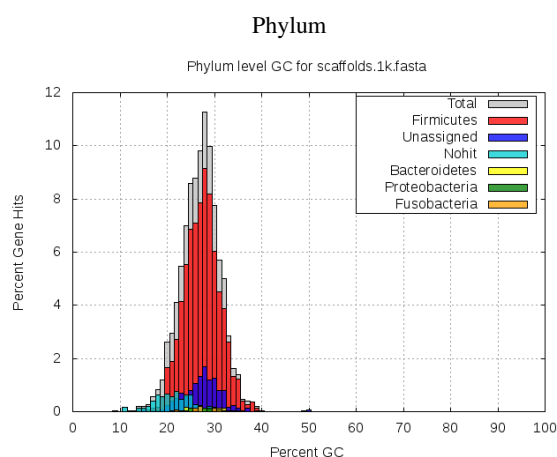
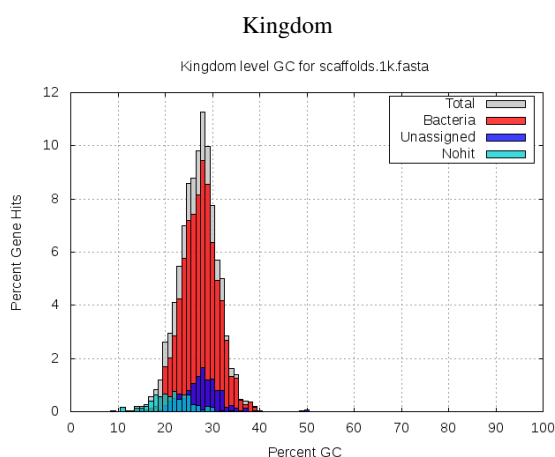


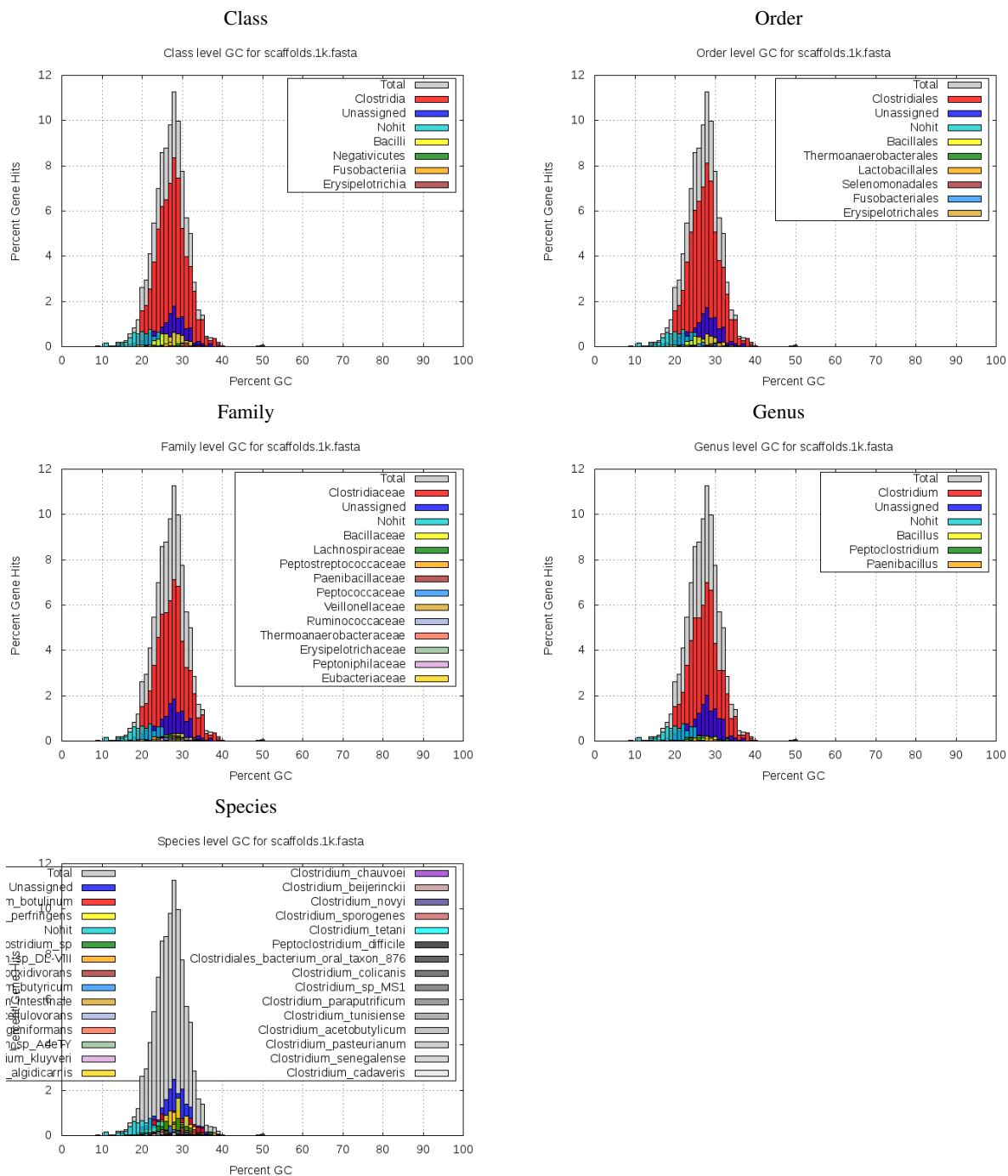
4. Assembly Statistics

Assembly method	SPAdes
Scaffold total	64
Contig total	64
Scaffold sequence length	2.747 Mb
Contig sequence length	2.747 Mb 0.000% gap
Scaffold N/L50	9/94.344 kb
Contig N/L50	9/94.344 kb
Largest Contig	303.3 kb
Number of scaffolds >50 kb	17
Pct of genome in scaffolds >50 kb	70.73

5. Assembly QC Results

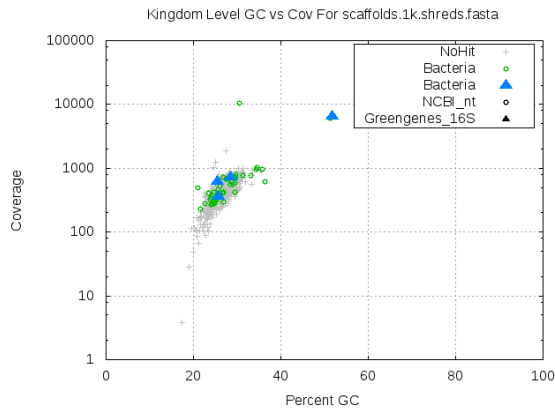
GC histogram of the predicted genes on each scaffold, overlaid with GC of hits based on LAST, shown for different taxonomic levels.



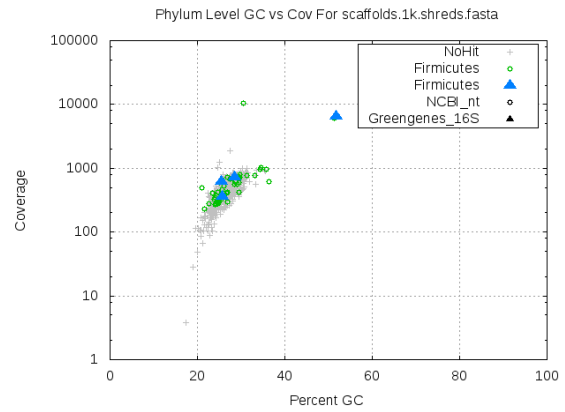


GC vs coverage based on GC of NCBI nt and Greengenes 16S rRNA gene hits to the assembly using megablast, shown for different taxonomic levels.

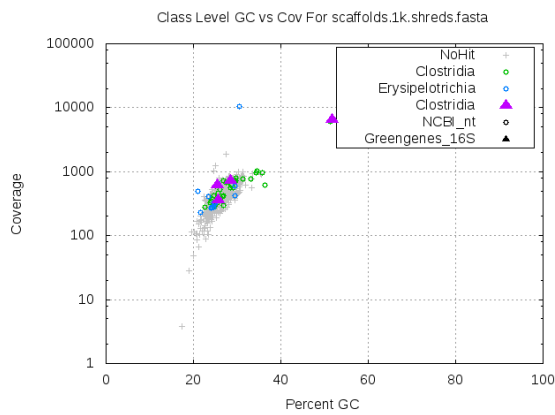
Kingdom



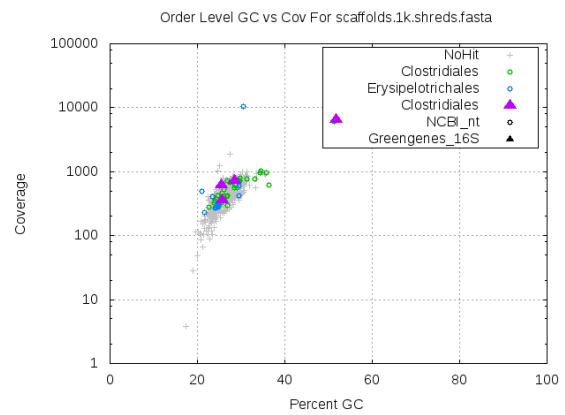
Phylum



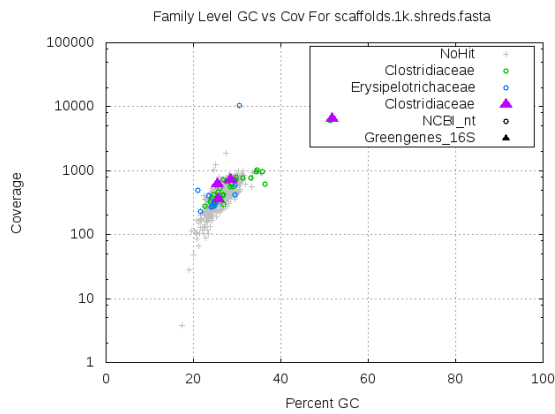
Class



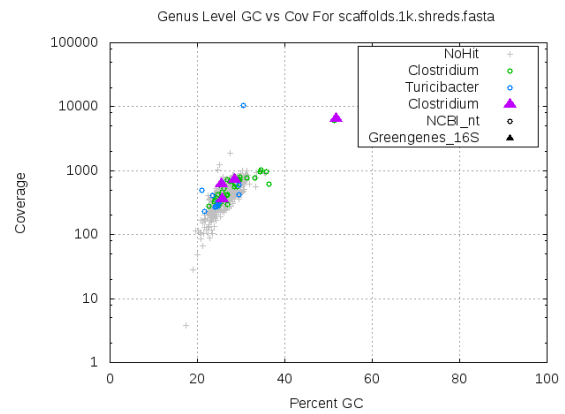
Order

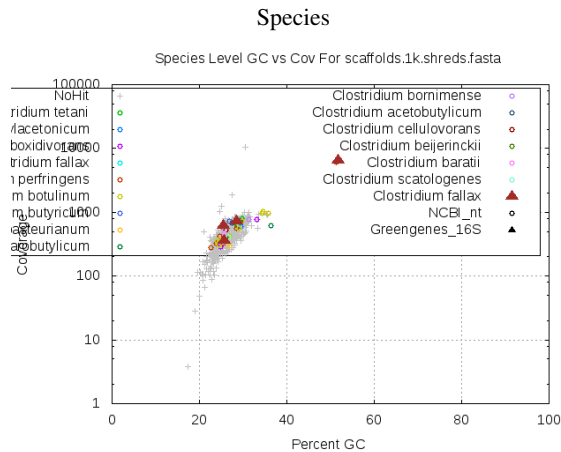


Family

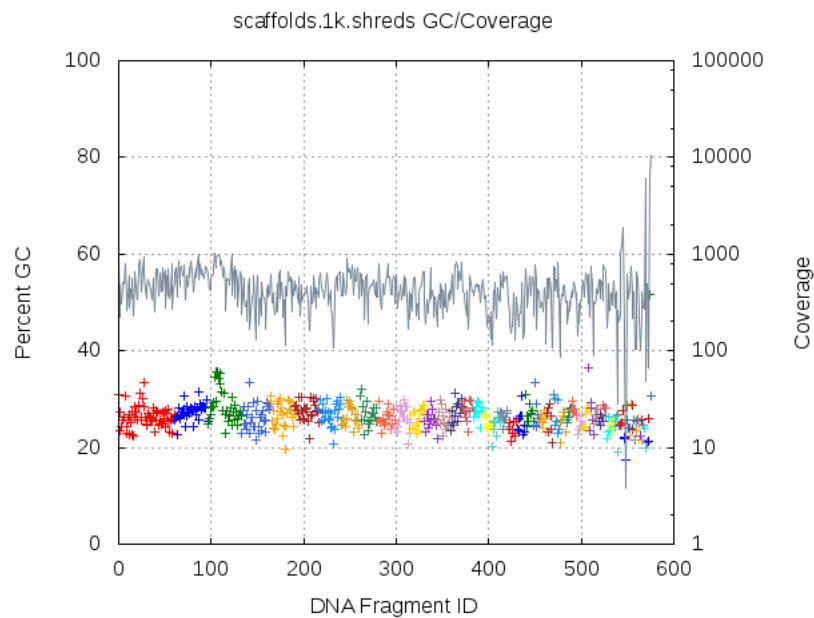


Genus

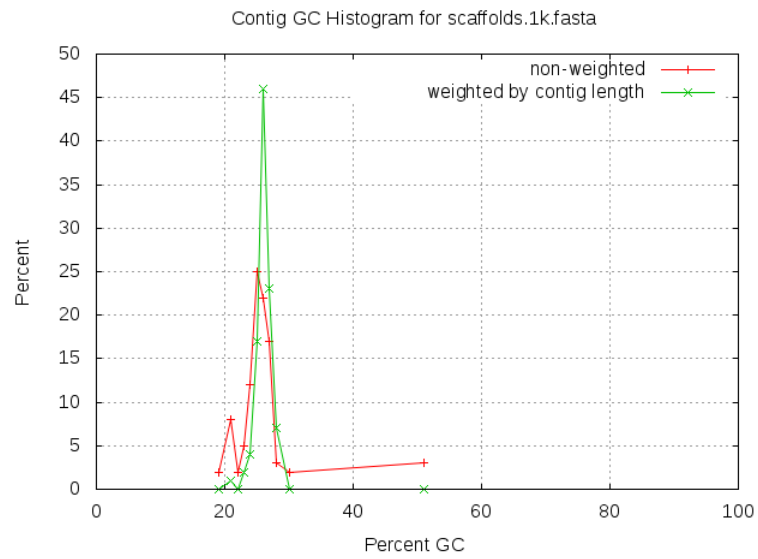




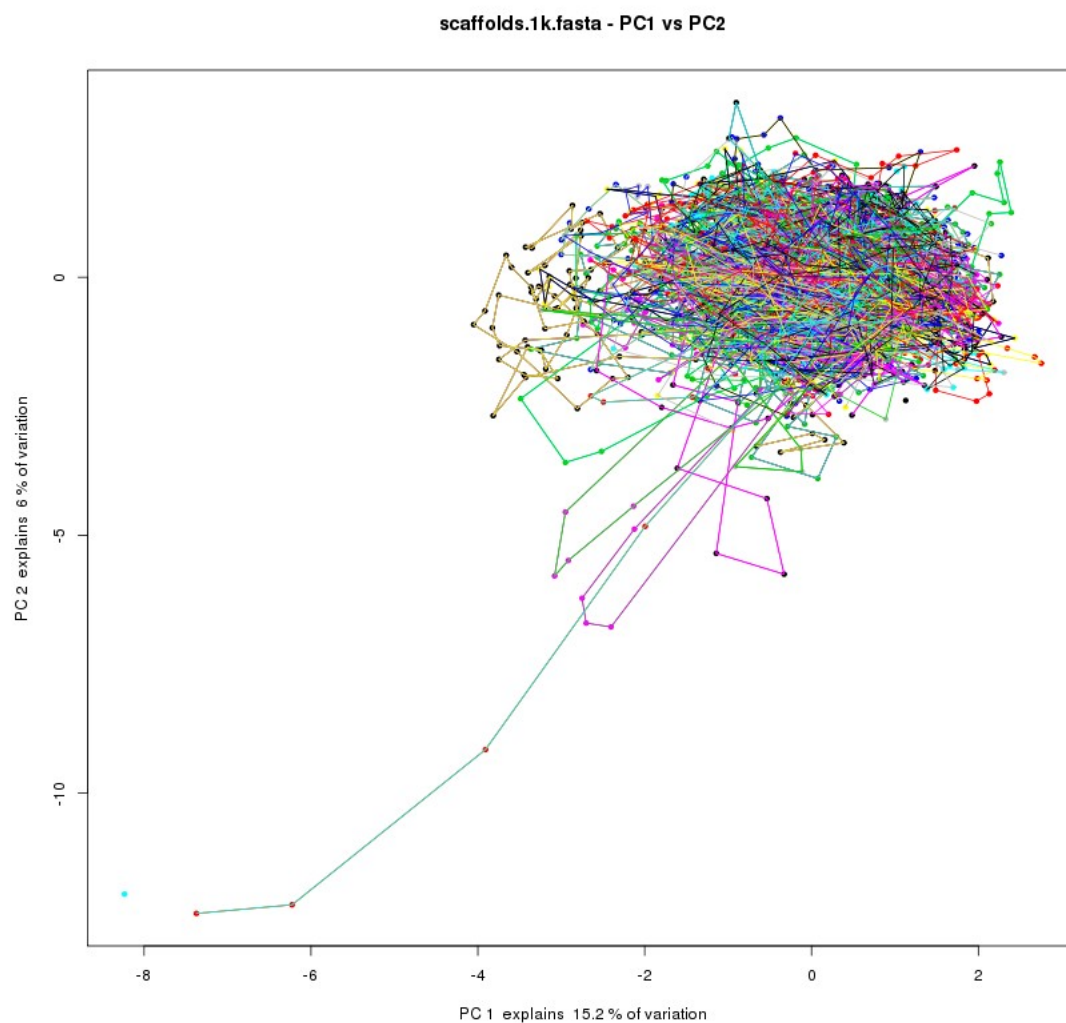
Coverage vs GC. Scaffolds were shredded into non-overlapping 5 kbp fragments and the GC content of each shred was plotted as a point, colored by scaffold id. Coverage was calculated by mapping the fragment library to the final assembly and plotted as connected points.



GC histogram of the scaffolds, including scaffold length weighted distribution.

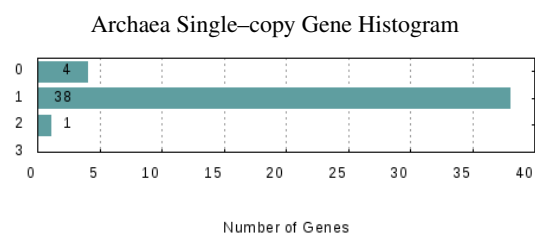
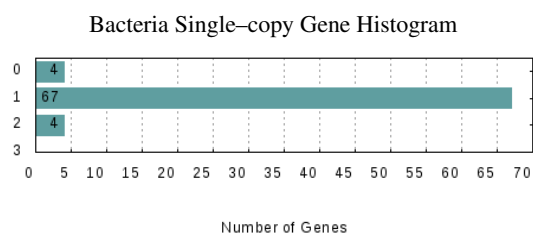


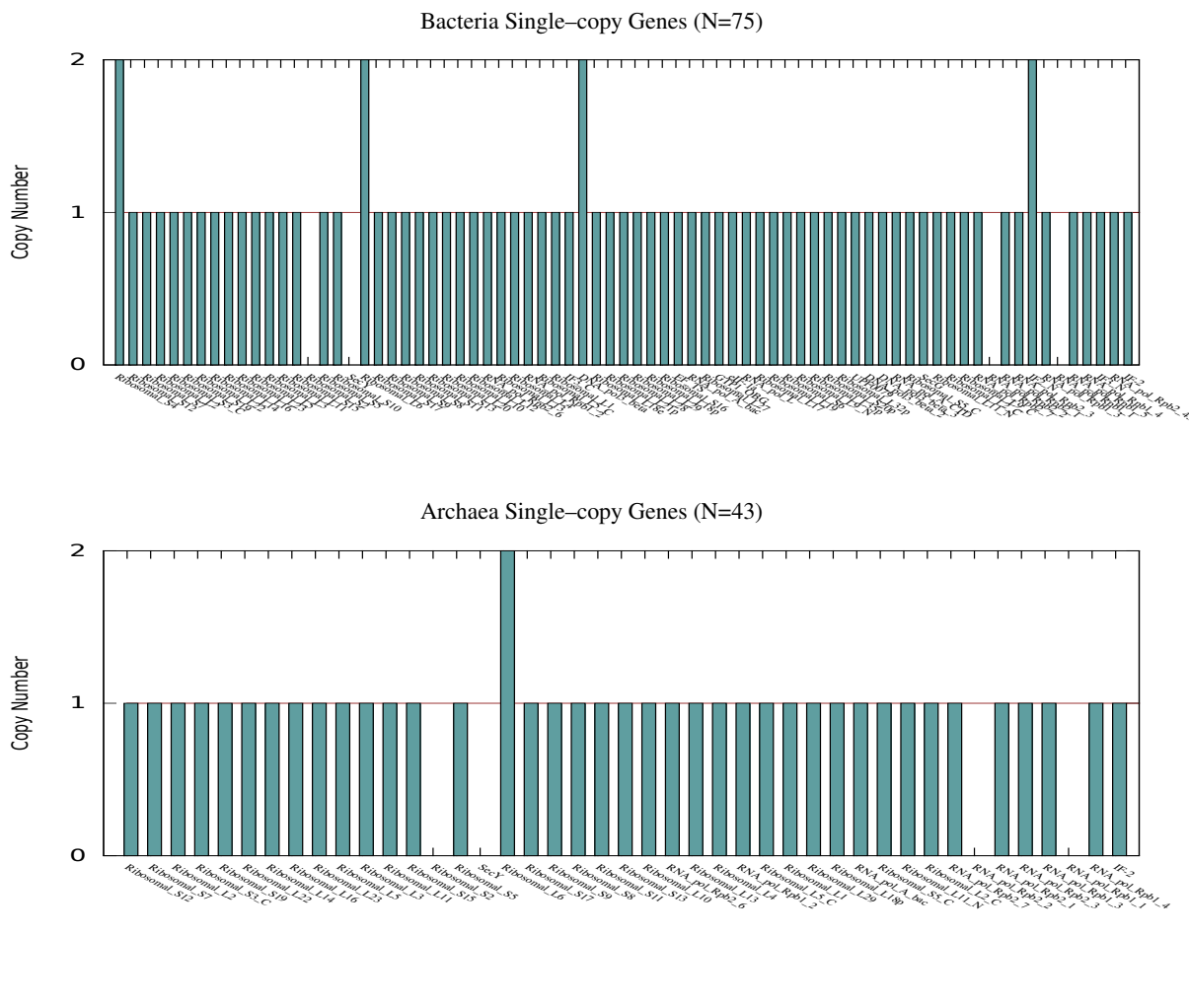
Principal component analysis of tetramer frequencies of scaffolds. Detectable variations are highlighted in color.



Estimated genome recovery derived from analysis of universal single-copy genes detected in final assembly.

HMM	Pct Recovered
archaea	100 %
bacteria	100 %





6. Sequence Data Availability

The following sequence fasta files can be downloaded from our JGI portal website.
<http://www.jgi.doe.gov/genome-projects>

7. Annotation Data Availability

The annotation of the assembled contigs can be found within IMG.
<http://img.jgi.doe.gov>

8. Methods

Isolate Minimal Draft

Genome sequencing and assembly

The draft genome of *Clostridium fallax* was generated at the DOE Joint Genome Institute (JGI) using the Illumina technology [1]. An Illumina 300 bp insert standard shotgun library was constructed and sequenced using the Illumina

HiSeq-2500 1TB platform which generated 8,789,838 reads totaling 1,309.7 Mbp. All general aspects of library construction and sequencing performed at the JGI can be found at <http://www.jgi.doe.gov>. All raw Illumina sequence data was filtered using BBDuk [2], which removes known Illumina artifacts and PhiX. Reads with more than one “N” or with quality scores (before trimming) averaging less than 8 or reads shorter than 51 bp (after trimming) were discarded. Remaining reads were mapped to masked versions of human, cat and dog references using BBMAP [2] and discarded if identity exceeds 95%. Sequence masking was performed with BBMask [2]. Following steps were then performed for assembly: (1) artifact filtered Illumina reads were assembled using SPAdes (version 3.6.2) [3]; (2) assembly contigs were discarded if length is <1 kbp. Parameters for the SPAdes assembly were `—cov-cutoff auto —phred-offset 33 -t 8 -m 40 —careful -k 25,55,95 —12`. The final draft assembly contained 64 contigs in 64 scaffolds, totalling 2.747 Mbp in size and was based on 1,309.7 Mbp of Illumina data with a mapped coverage of 488.7X.

1. Bennett S. Solexa Ltd. Pharmacogenomics. 2004;5(4):433–8.
 2. B. Bushnell: BBTools software package, URL <http://sourceforge.net/projects/bbmap>.
 3. Bankevich A, et.al, SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J Comput Biol 2012; 19:455–77.
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